AWS VPC PEERING
PROJECT REPORT

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Batch: 1

Program: EWB Cloud Labs

Region: Asia Pacific (Mumbai), ap-south-1

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Secure private communication between EC2 instances hosted in two separate Amazon VPCs using VPC Peering, route tables, security groups, and SSH ProxyJump.

1. Objective

Configure two non-overlapping VPCs and validate bidirectional private connectivity. Use a public bastion as an SSH jump host without copying the private SSH key to any EC2 instance. Capture failure before peering, success after routing, and controlled ICMP blocking/recovery.

2. Architecture

Laptop (WSL SSH client) -> Bastion (VPC A public subnet) -> App-A (VPC A private subnet) / App-B (VPC B private subnet). App-A and App-B communicate through VPC Peering.

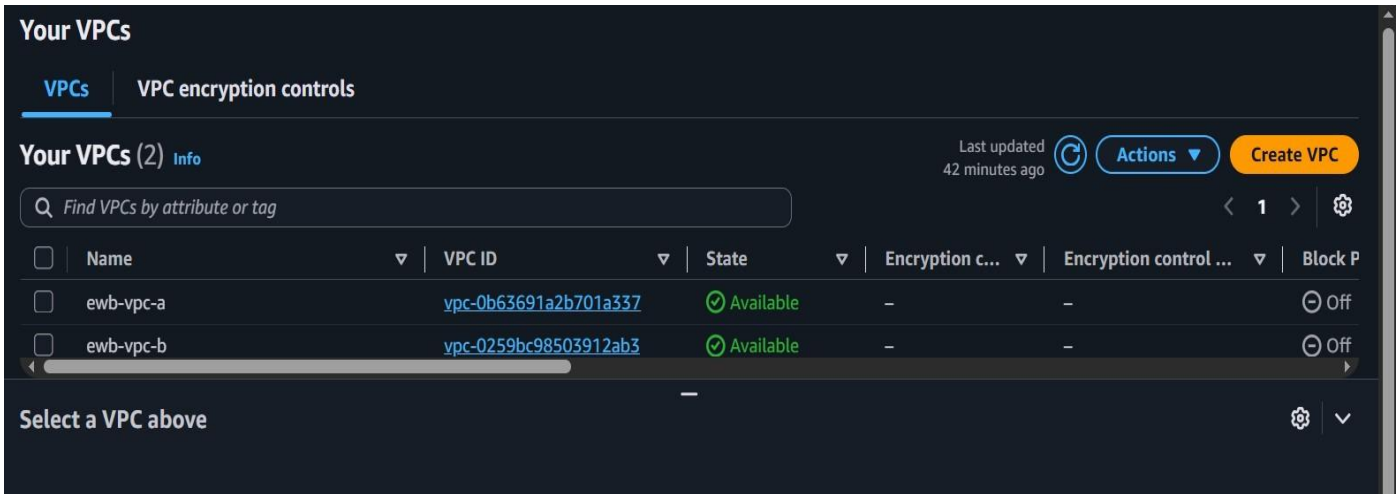
Component	Network / IP
VPC A	10.10.0.0/16
Public subnet A	10.10.1.0/24
Private subnet A	10.10.2.0/24
VPC B	10.20.0.0/16
Private subnet B	10.20.1.0/24
Bastion	10.10.1.248 (private); public IP enabled
App-A	10.10.2.162 (private only)
App-B	10.20.1.247 (private only)

Note: The actual EC2 private IPs differ from the workbook examples. Connectivity testing and rules use the actual instance IPs.

3. Resource inventory

Resource	AWS ID
VPC A	vpc-0b63691a2b701a337
VPC B	vpc-0259bc98503912ab3
Peering	pcx-0215add6339f8840b
Bastion	i-05ef7d8a5c92e2f69
App-A	i-0b6f31f73eb87d2f8
App-B	i-05db79c322526d403

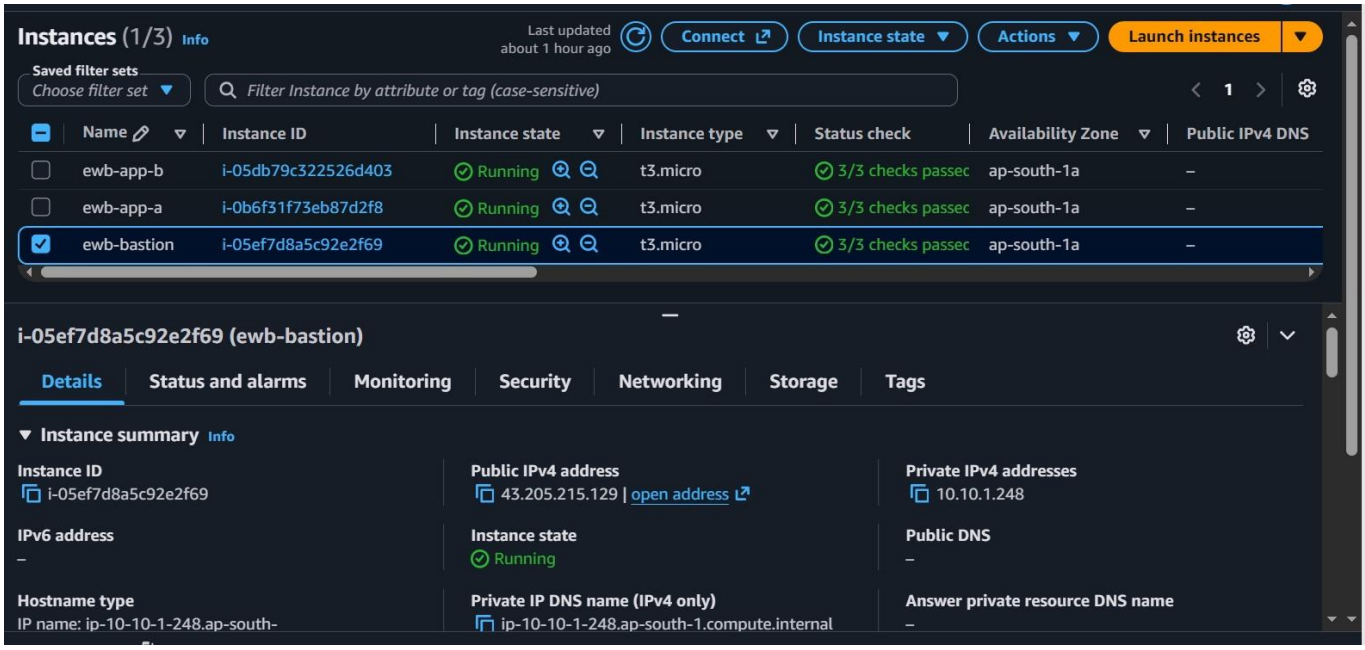
Evidence: both VPCs are available.



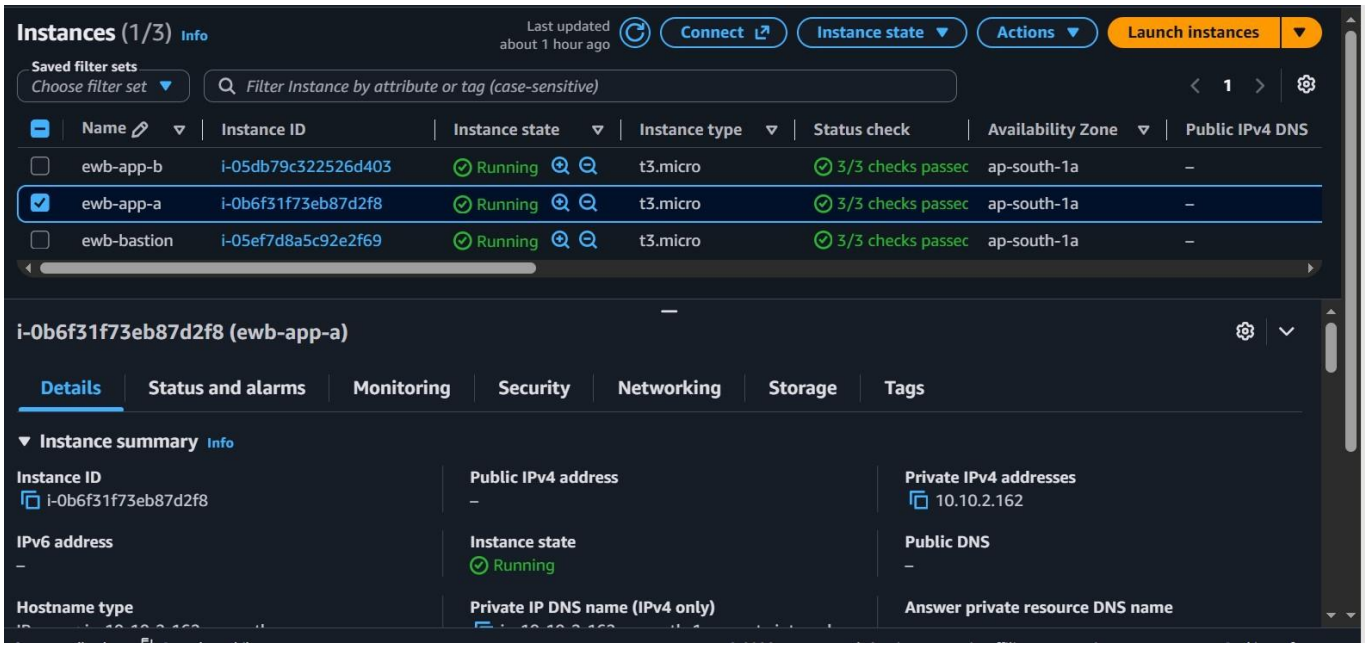
4. EC2 instances

All three instances use Amazon Linux 2023 and t3.micro, in ap-south-1a. App-A and App-B have no public IPv4 address.

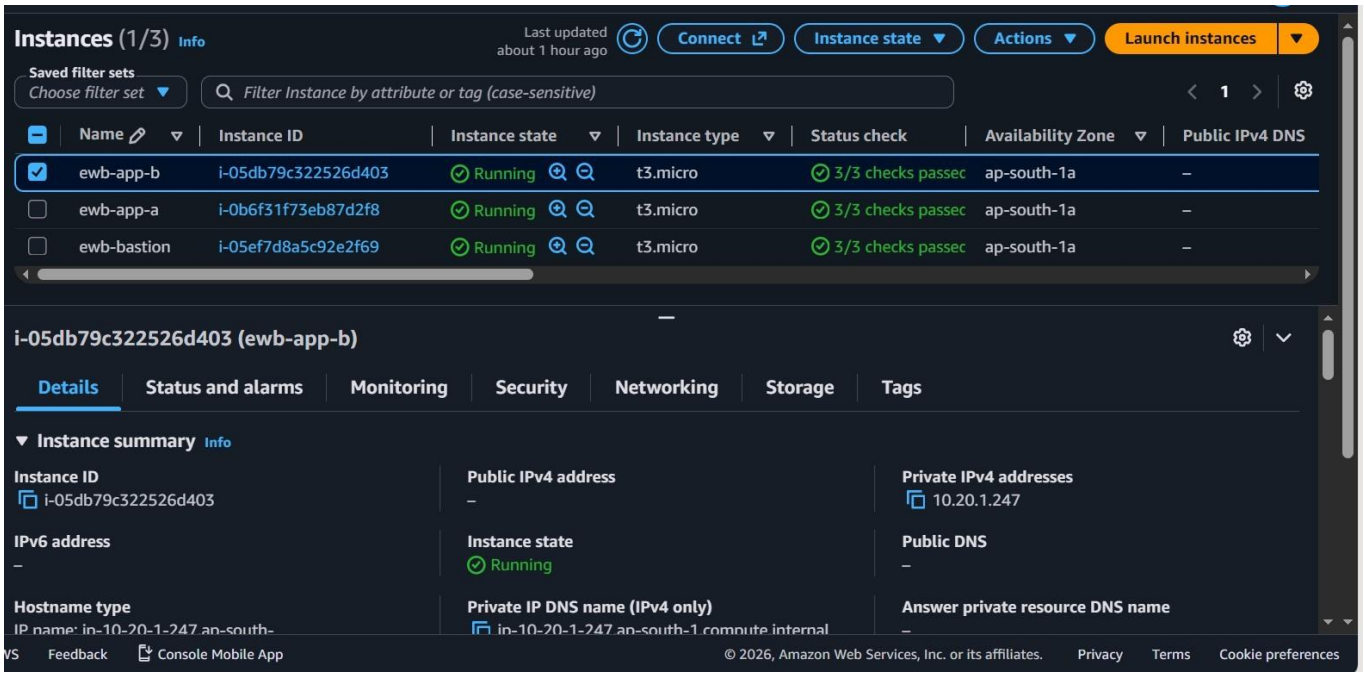
Bastion: public and private IPv4 addresses.



App-A: private IPv4 only.



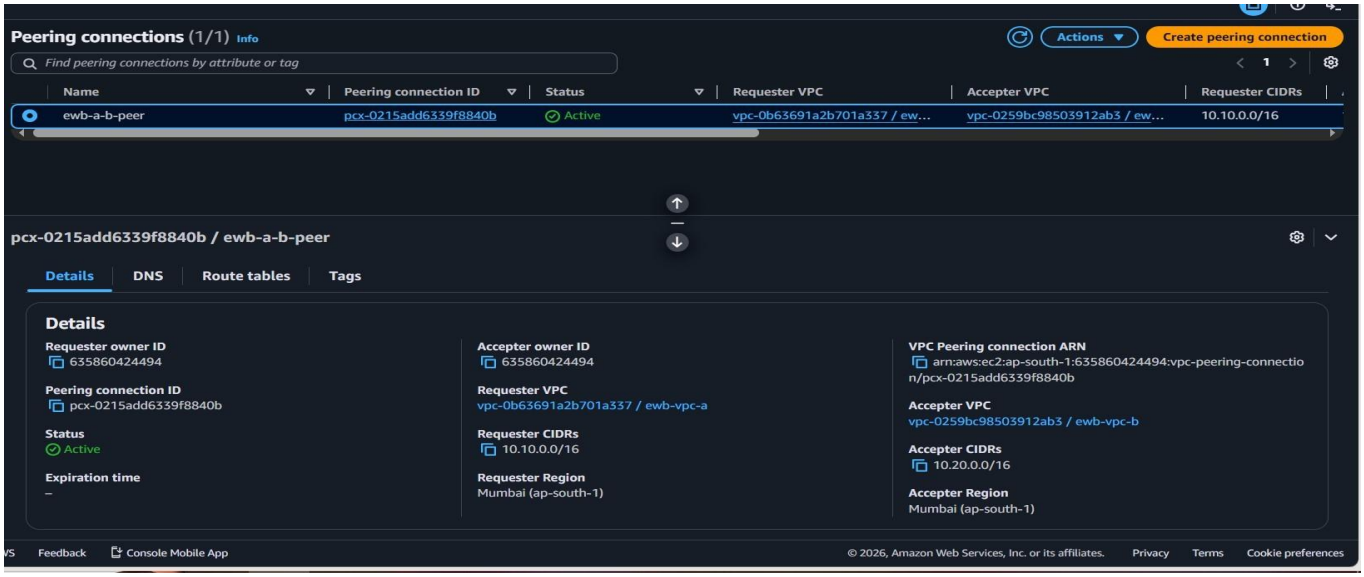
App-B: private IPv4 only.



5. Peering connection

Requester: VPC A; accepter: VPC B. Both in the Mumbai Region and the same AWS account. Connection is Active.

Evidence: Active peering connection and both VPC IDs.

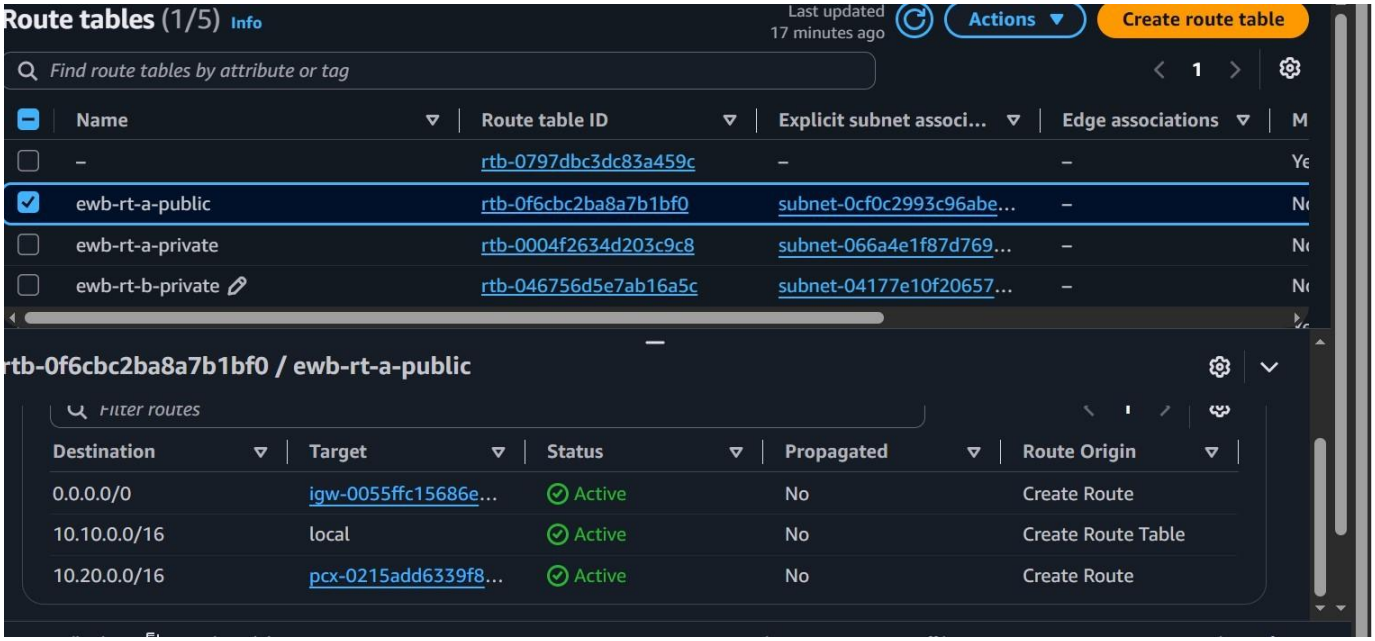


6. Route table configuration

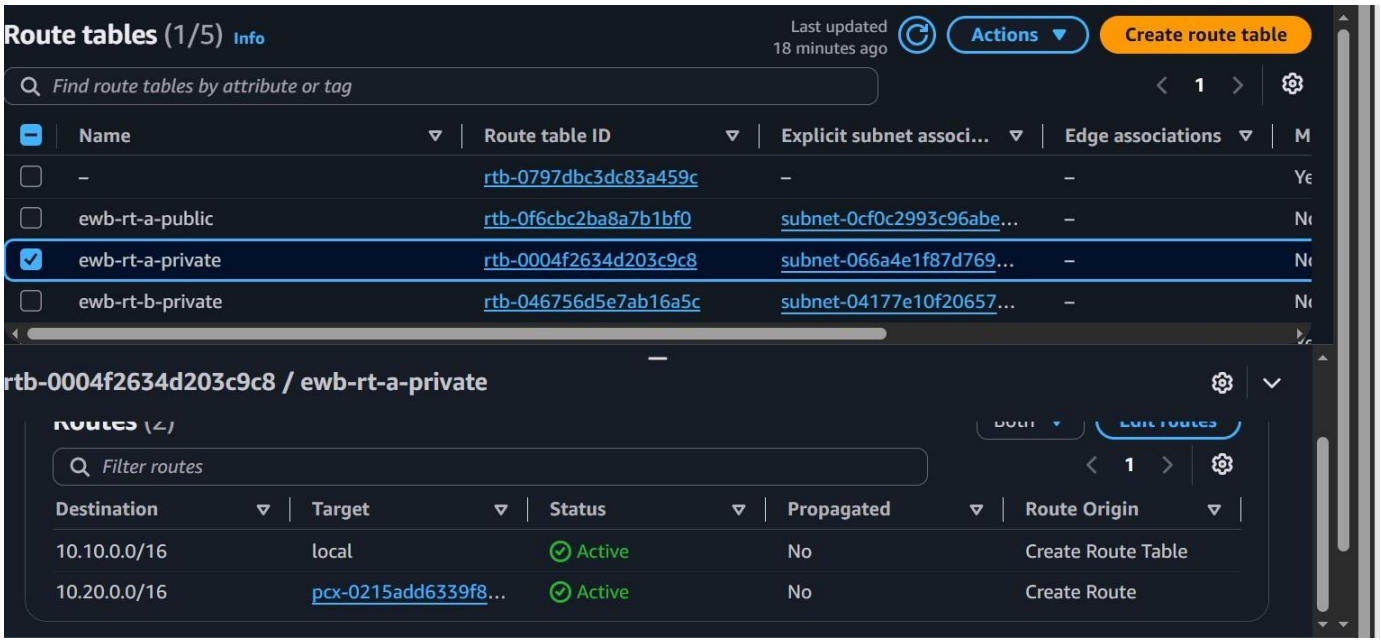
Route table	Destination	Target
ewb-rt-a-public	0.0.0.0/0; 10.20.0.0/16	Internet Gateway; Peering
ewb-rt-a-private	10.20.0.0/16	Peering
ewb-rt-b-private	10.10.0.0/16	Peering

Local VPC routes remain present. Route tables are explicitly associated with their intended subnets.

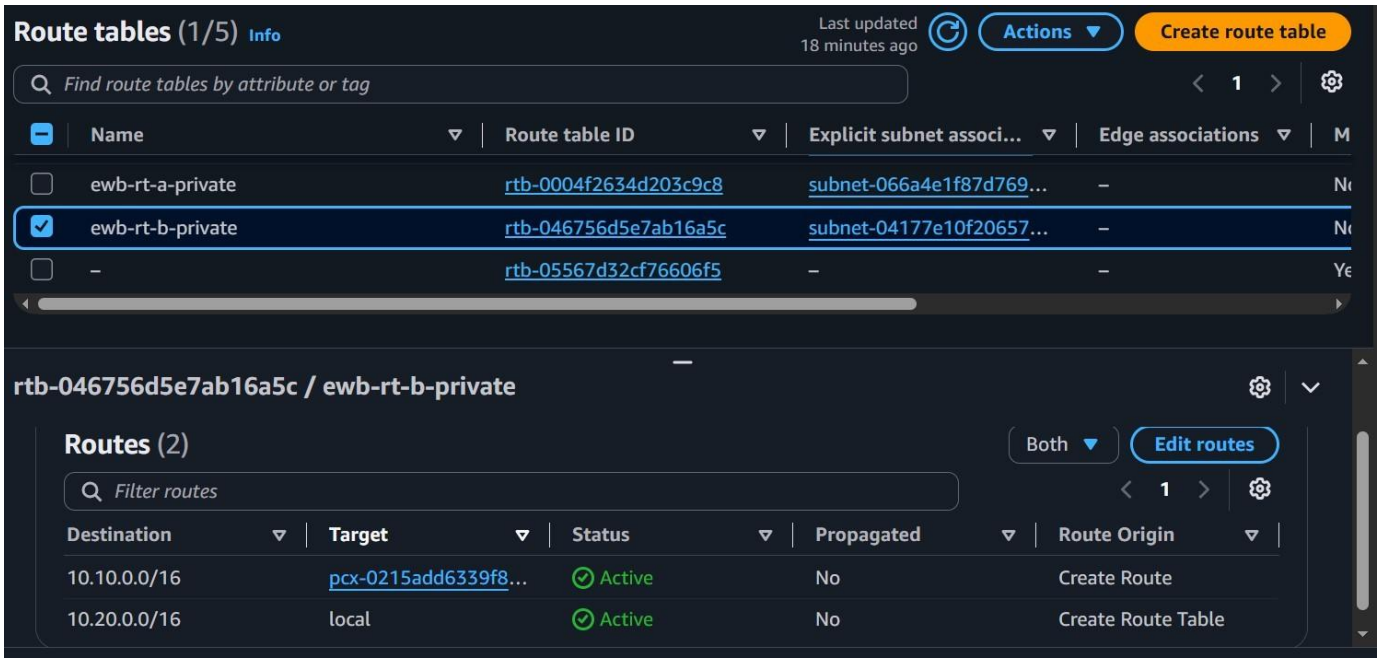
VPC A public route table.



VPC A private route table.



VPC B private route table.

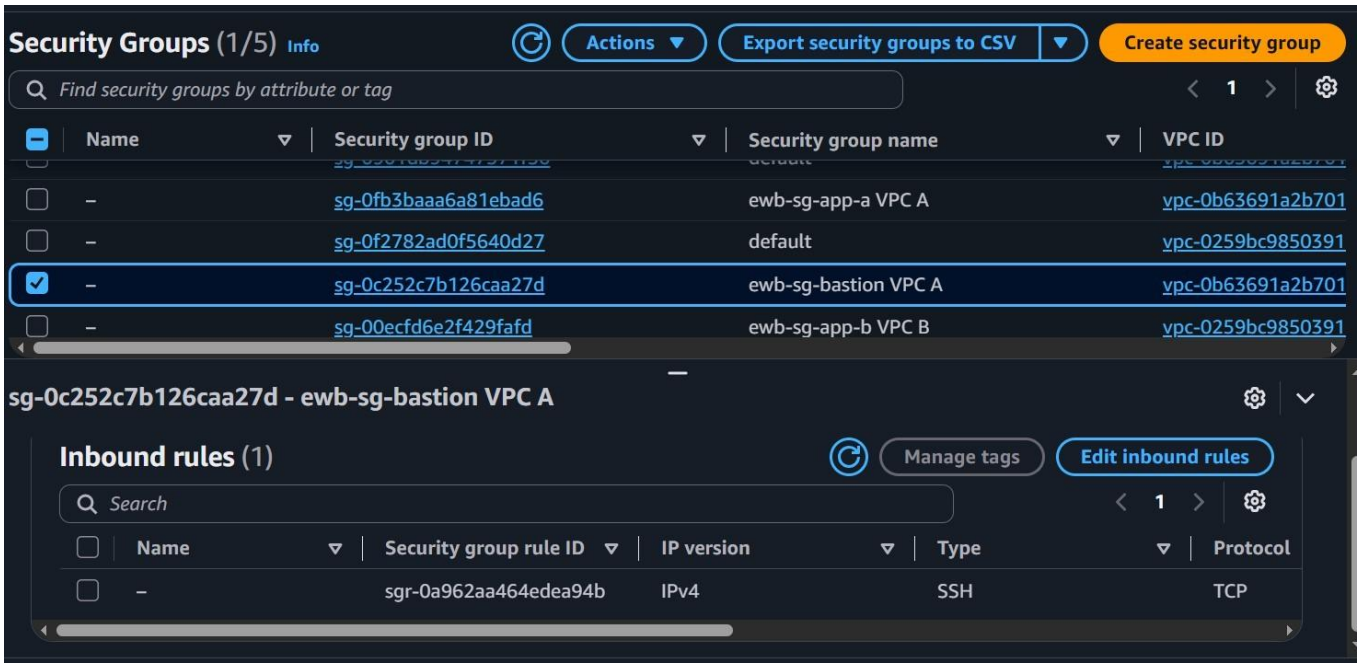


7. Security groups

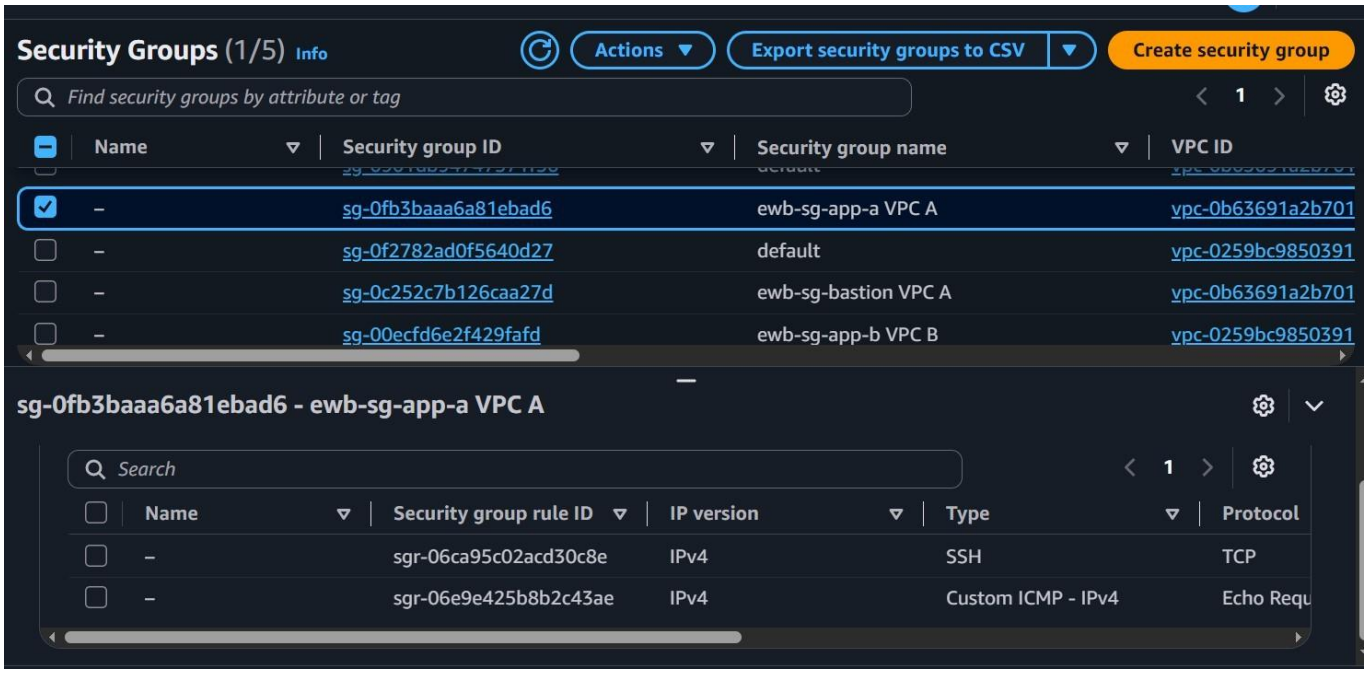
Security group	Inbound permissions
Bastion	SSH 22 from laptop public IPv4 /32
App-A	SSH 22 from 10.10.1.248/32; ICMP Echo Request from 10.20.1.247/32
App-B	SSH 22 from 10.10.1.248/32; ICMP Echo Request from 10.10.2.162/32

Inbound source values are recorded from the working configuration; the screenshots below show rule types but not the complete source column.

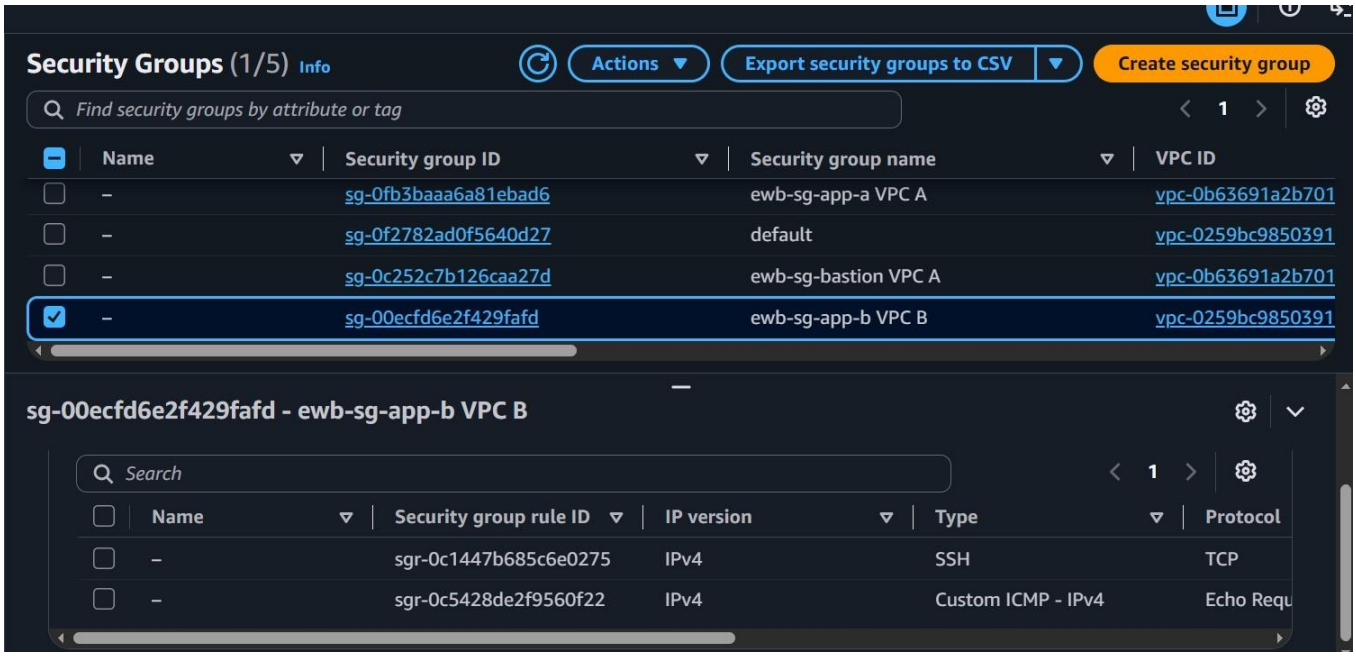
Bastion security group.



App-A security group.



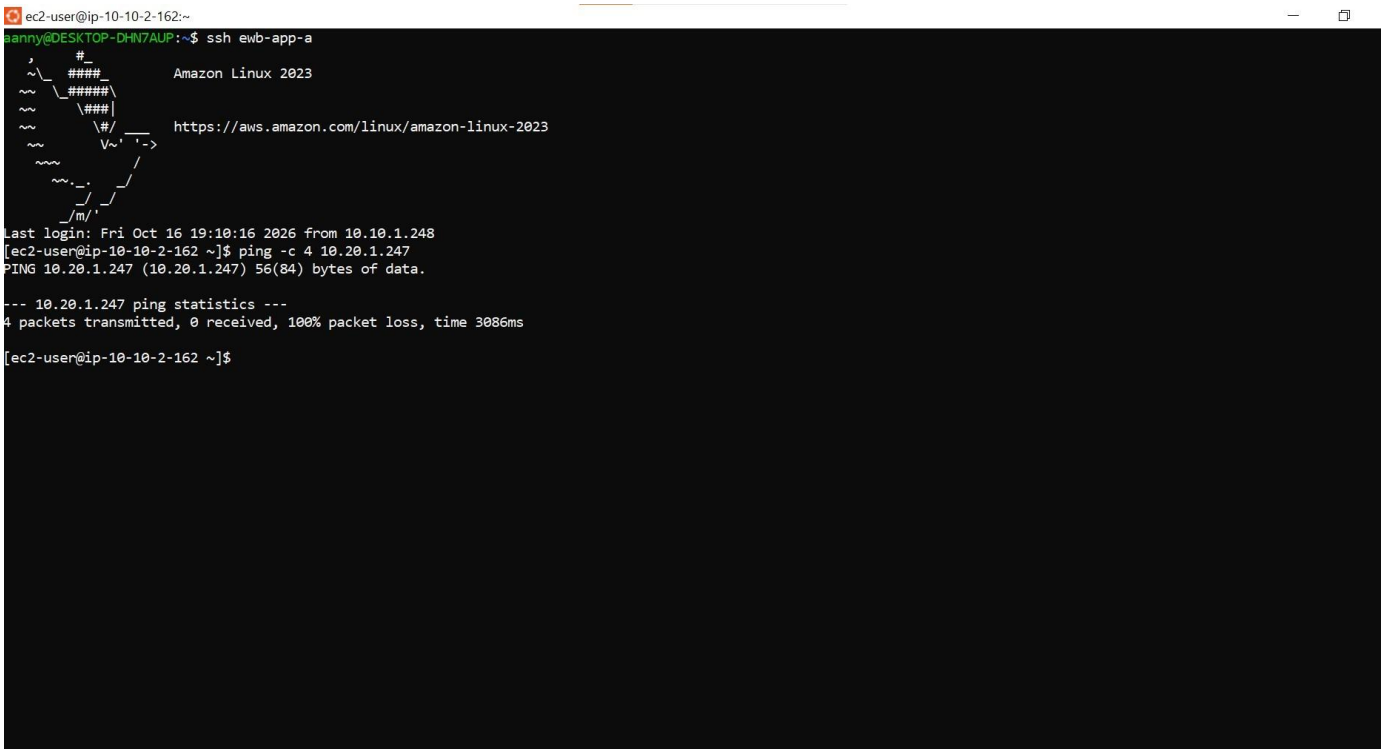
App-B security group.



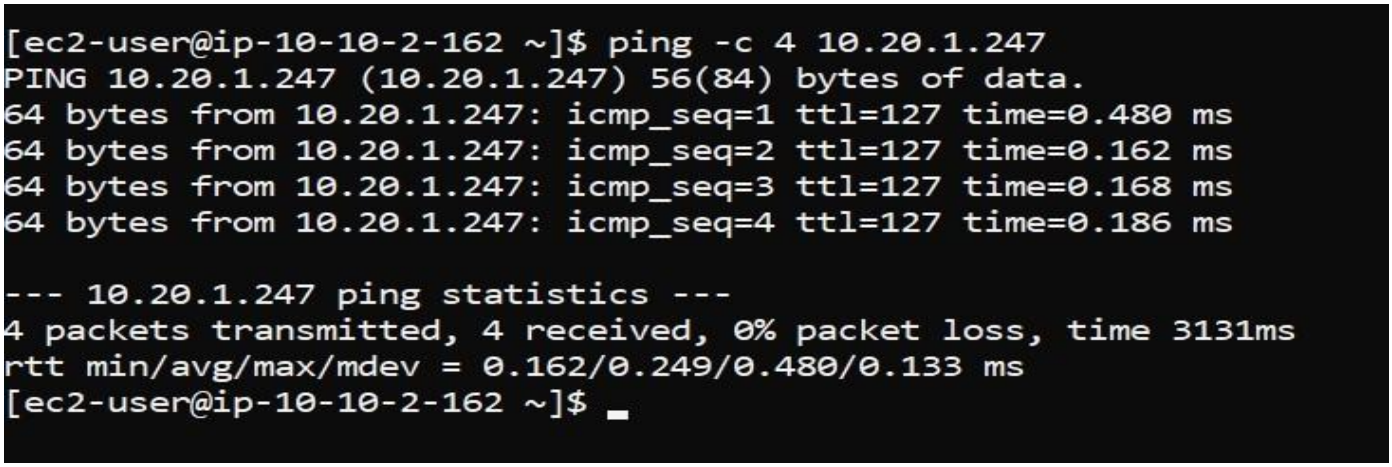
8. Connectivity and validation

SSH ProxyJump was used from WSL to connect to App-A and App-B through Bastion. The private SSH key remained on the laptop.

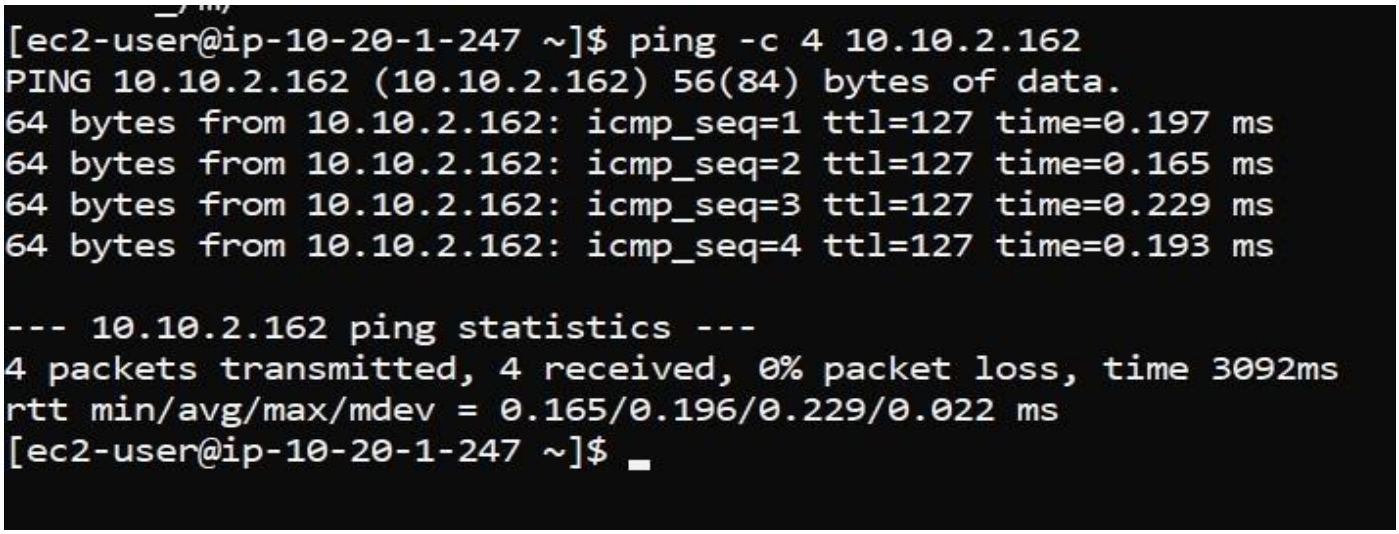
Baseline: App-A to App-B before peering; 100% packet loss.



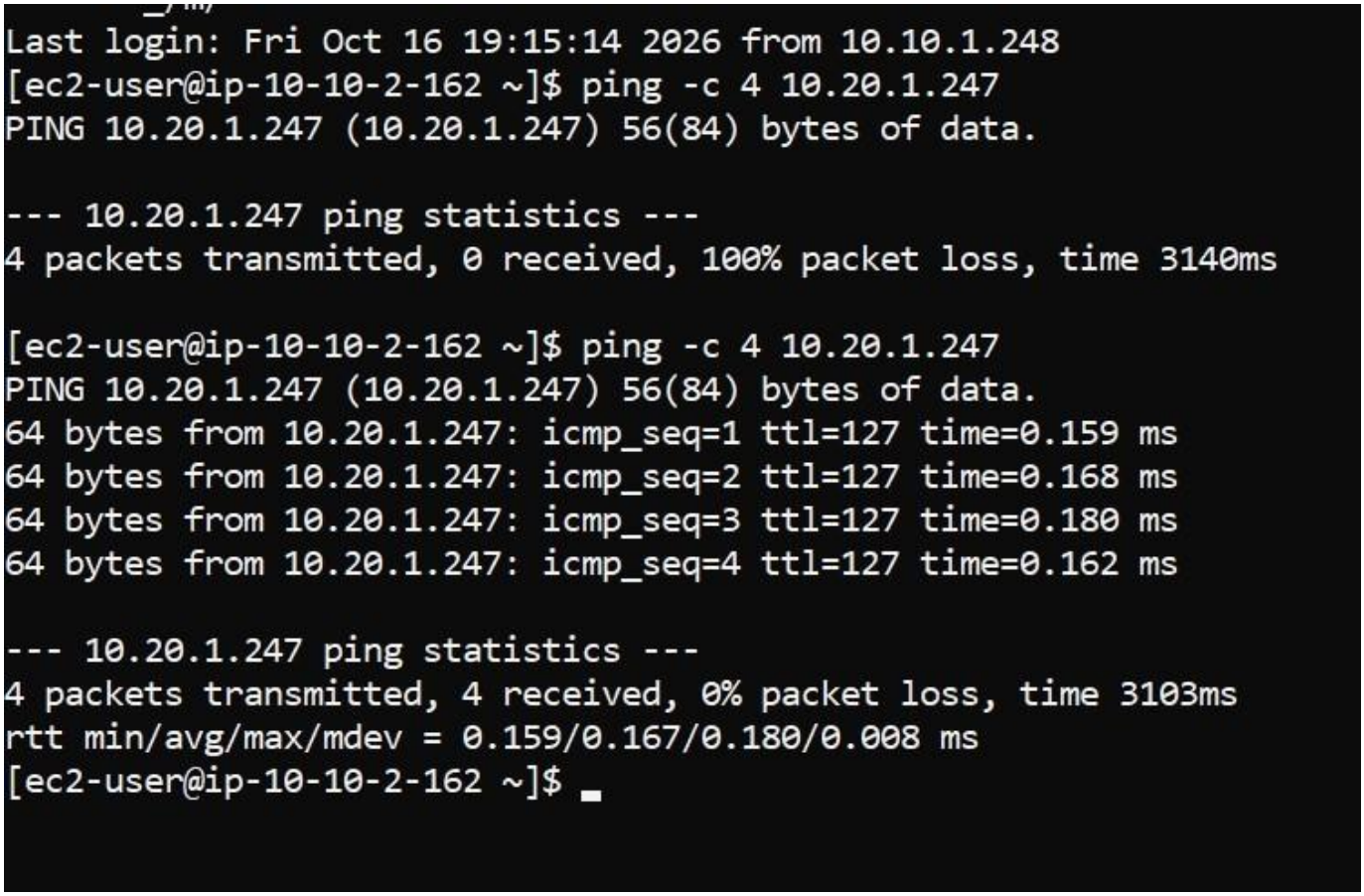
After peering: App-A to App-B; 0% packet loss.



After peering: App-B to App-A; 0% packet loss.



Controlled failure and recovery: ICMP blocked (100% loss), then restored (0% loss).



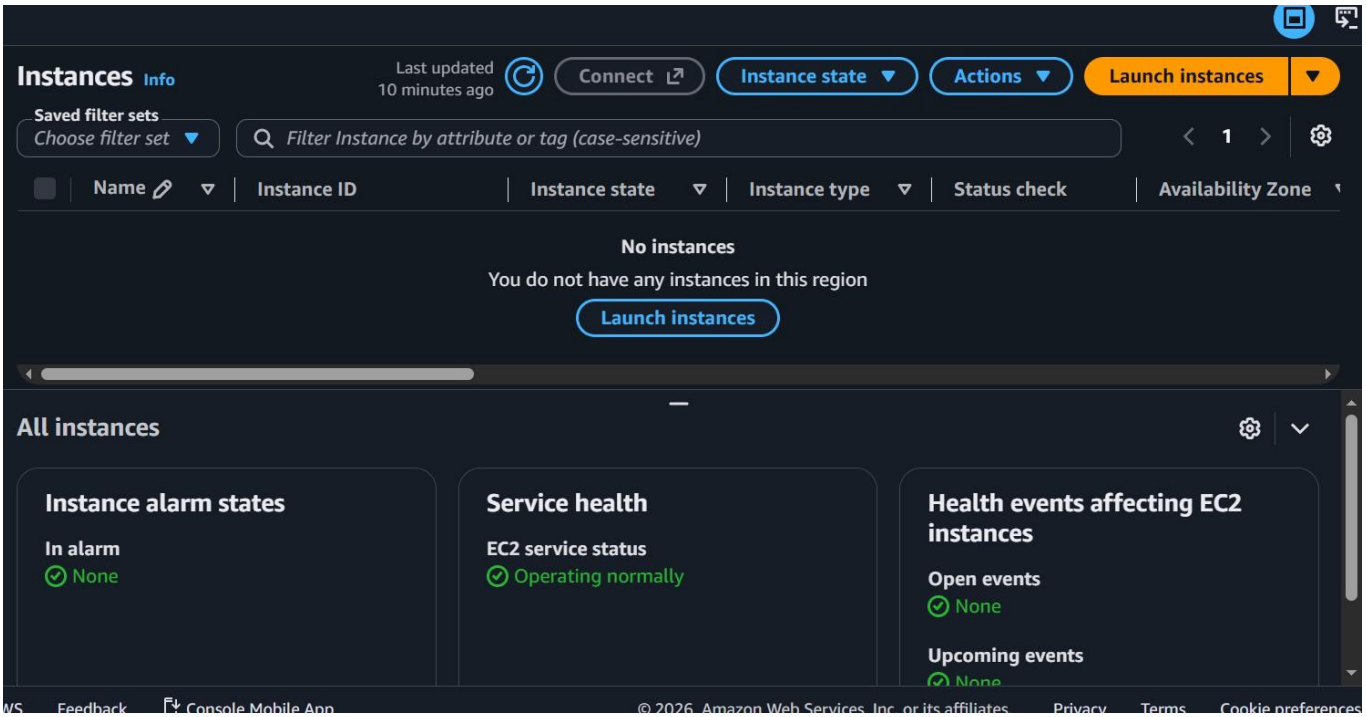
9. Lessons learned

VPC Peering provides a private path between non-overlapping VPC CIDR ranges, but routing and security group permissions must also be configured. Security groups control allowed inbound traffic; route tables determine the next hop. SSH ProxyJump avoids copying sensitive private keys to the bastion.

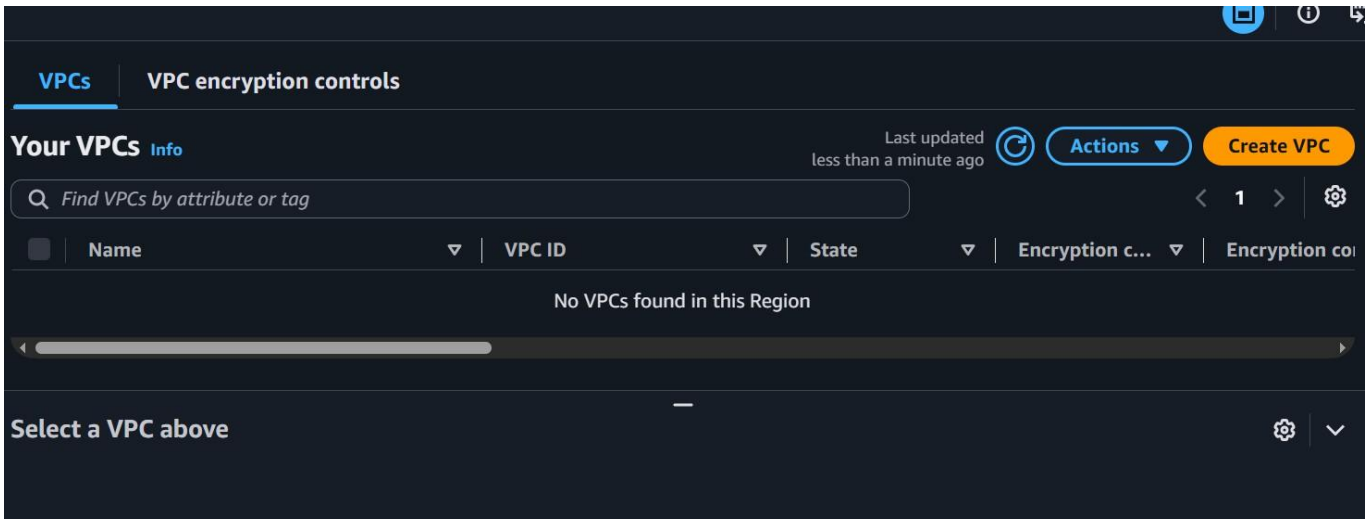
10. AWS Resource Cleanup Evidence

After connectivity testing, the project resources were removed. The following screenshots show the AWS Console state after cleanup. These screenshots confirm that no instances, VPCs, or peering connections are listed in the selected Region; they do not record the individual deletion actions.

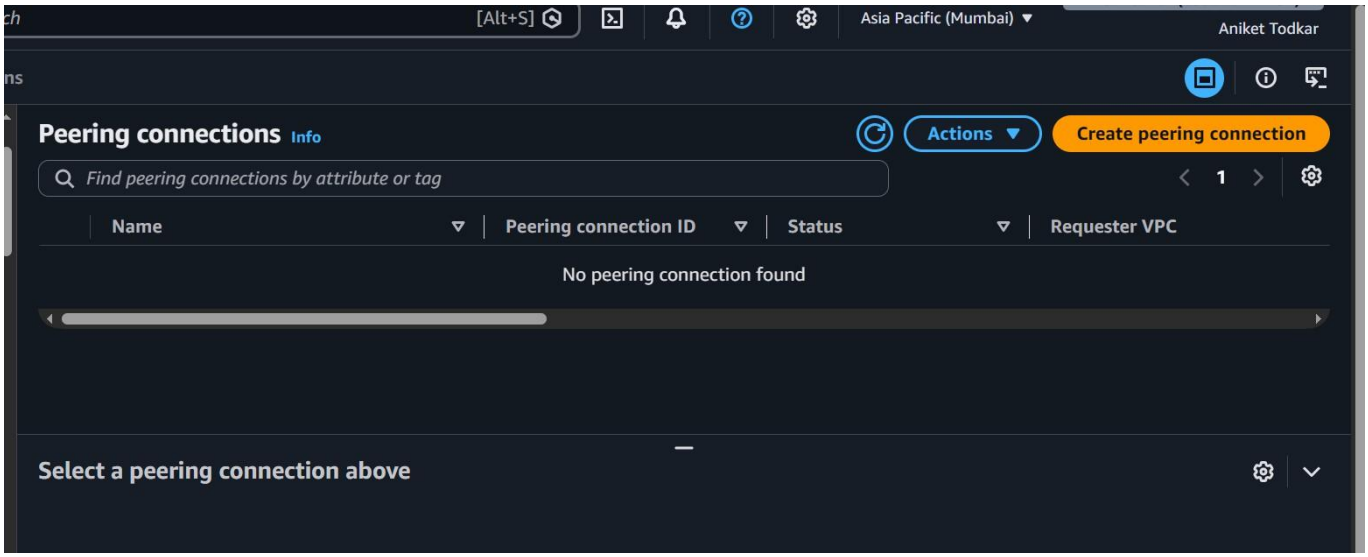
Evidence 1 | Host / Console: AWS EC2 Instances
Test: EC2 cleanup verification
Result: No instances found in the selected Region



Evidence 2 | Host / Console: AWS VPC > Your VPCs
Test: VPC cleanup verification
Result: No VPCs found in the selected Region



Evidence 3 | Host / Console: AWS VPC > Peering Connections
Test: VPC peering cleanup verification
Result: No peering connection found (Mumbai Region)



Security: No private SSH keys, passwords, or access tokens are included.

11.Forward Path and Return Path

Forward (ICMP Echo Request): App-A (10.10.2.162) -> VPC A private subnet route table (10.20.0.0/16 via peering) -> VPC peering connection -> VPC B -> App-B (10.20.1.247).

Return (ICMP Echo Reply): App-B (10.20.1.247) -> VPC B private subnet route table (10.10.0.0/16 via peering) -> VPC peering connection -> VPC A -> App-A (10.10.2.162).

Both directions require appropriate subnet route entries and security group permissions. Peering does not automatically add routes or allow inbound traffic.

12. Connectivity Screenshot Labels

Host	Test	Result
App-A	Ping App-B before peering	FAIL - 100% packet loss
App-A	Ping App-B after peering	PASS - 0% packet loss
App-B	Ping App-A after peering	PASS - 0% packet loss
App-A	Ping App-B with ICMP blocked	FAIL - 100% packet loss
App-A	Ping App-B after ICMP restored	PASS - 0% packet loss

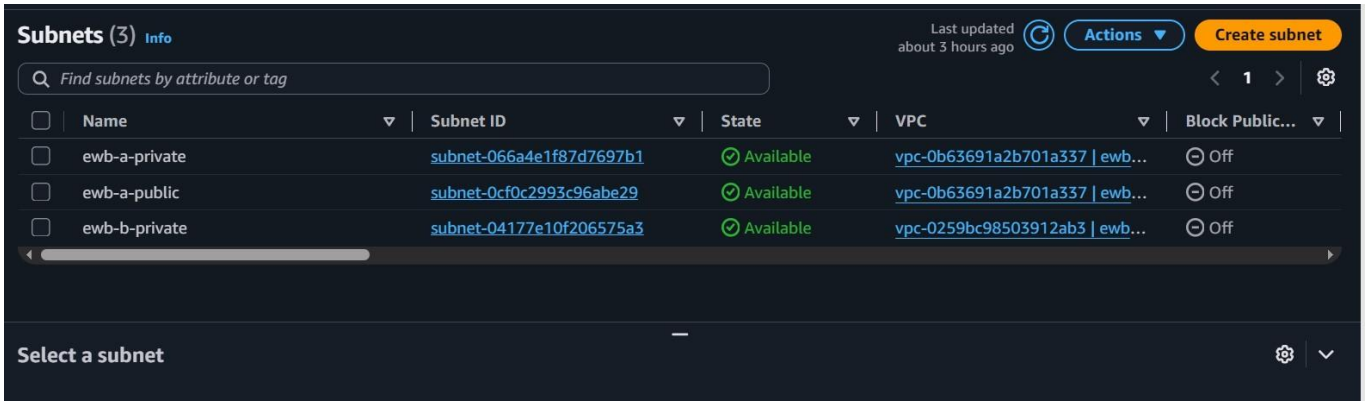
13. What I Learned

I learned to configure two VPCs with non-overlapping CIDRs, route traffic through a VPC peering connection, restrict access using security groups, and test connectivity using SSH ProxyJump and ICMP. The controlled failure test demonstrated that connectivity depends on both routing and security rules. I also learned to remove lab resources after validation to reduce unnecessary AWS costs.

Security note: No private keys, passwords, or access tokens are included in this report.

14. Appendix: Additional Configuration Screenshots

Console: VPC Subnets | **Test:** Verify three lab subnets | **Result:** ewb-a-public, ewb-a-private and ewb-b-private shown as Available.



15. Conclusion

The AWS VPC Peering project demonstrated private communication between two VPCs with non-overlapping CIDR blocks. Connectivity testing showed that App-A could not reach App-B before the required peering routes were configured, but bidirectional ICMP tests succeeded after configuration. Removing the relevant ICMP inbound rule caused a controlled failure, and restoring it recovered connectivity.
SSH access through the Bastion host using ProxyJump provided administration without assigning public IP addresses to the application instances.

16. References

1. EWB Cloud Labs, AWS VPC Peering - Student Project Workbook, course-provided project instructions.

2. AWS Documentation, What is VPC peering?
   https://docs.aws.amazon.com/vpc/latest/peering/what-is-vpc-peering.html

3. AWS Documentation, Update your route tables for a VPC peering connection.
   https://docs.aws.amazon.com/vpc/latest/peering/vpc-peering-routing.html

4. AWS Documentation, Security groups for your VPC.
   https://docs.aws.amazon.com/vpc/latest/userguide/vpc-security-groups.html

5. OpenSSH Manual, ssh_config (ProxyJump directive).
   https://man.openbsd.org/ssh_config